SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)

Project Report on

"Automatic Railway Crossing"

submitted in partial fulfillment of the requirement for the completion of V semester of

BACHELOR OF ENGINEERING

in

ELECTRONICS & TELECOMMUNICATION ENGINEERING

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DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION ENGINEERING

CERTIFICATE

Certified that the mini project work entitled "Automatic Railway Crossing" is a bonafide work carried out by Harshitha N (1SI23ET016), Keerthana M K (1SI23ET021), Lavanya G S (1SI23ET024) and Mamatha (1SI23ET028) in partial fulfillment for the completion of V Semester of Bachelor of Engineering in Electronics & Telecommunication Engineering from Siddaganga Institute of Technology, an autonomous institute under Visvesvaraya Technological University, Belagavi during the academic year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The Mini project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

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Course Outcomes

CO1: To identify a problem through literature survey and knowledge of contemporary engineering technology.

CO2: To consolidate the literature search to identify issues/gaps and formulate the engineering problem

CO3: To prepare project schedule for the identified design methodology and engage in budget analysis, and share responsibility for every member in the team

CO4: To provide sustainable engineering solution considering health, safety, legal, cultural issues and also demonstrate concern for environment

CO5: To identify and apply the mathematical concepts, science concepts, engineering and management concepts necessary to implement the identified engineering problem

CO6: To select the engineering tools/components required to implement the proposed solution for the identified engineering problem

CO7: To analyze, design, and implement optimal design solution, interpret results of experiments and draw valid conclusion

CO8: To demonstrate effective written communication through the project report, the one-page poster presentation, and preparation of the video about the project and the four page IEEE/Springer/ paper format of the work

CO9: To engage in effective oral communication through power point presentation and demonstration of the project work

CO10:To demonstrate compliance to the prescribed standards/ safety norms and abide by the norms of professional ethics

CO11: To perform in the team, contribute to the team and mentor/lead the team

POs: PO1: Engineering Knowledge, PO2: Problem analysis, PO3: Design/Development of solutions, PO4: Conduct investigations of complex problems, PO5: Modern tool usage, PO6: Engineer and the world, PO7: Ethics, PO8: Individual and collaborative team work, PO9: Communication, PO10: Project management and finance, PO11: Lifelong learning,PSO1:Apply and analyze the concepts of circuits and systems,PSO2:Identify, design and develop solutions for complex engineering problems

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO-1											3		
CO-2		3											
CO-3											3		
CO-4						3	3						
CO-5	3	3											
CO-6					3								
CO-7			3	3									
CO-8										3			
CO-9										3			
CO-10								3					
CO-11									3				

Attainment level: - 1: Slight (low) 2: Moderate (medium) 3: Substantial (high)

Abstract

Railway level crossings are critical points of interaction between railways and road traffic, where improper control can lead to severe accidents and traffic congestion. Manual operation of railway crossing gates is often prone to human error and may not be feasible in unmanned or remote locations. This project presents the design and implementation of an automatic railway crossing system that operates without human intervention to enhance safety and reliability.

The proposed system automatically identifies an approaching train, determines its speed, and controls the opening and closing of the gate with appropriate timing. By controlling the gate operation based on train movement, the system ensures that the gate is securely closed before the train arrives and reopened only after the train has completely passed. Visual indications and real-time status messages are provided to inform road users about the gate condition.

The system reduces dependence on manual control, minimizes accidents caused by negligence, and decreases unnecessary waiting time for road traffic. It offers a cost-effective, and reliable solution for modern railway crossings, especially in areas with limited manpower. The successful implementation demonstrates the potential of automation in improving safety, traffic management, and operational efficiency at railway level crossings.

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Chapter 1

Introduction

Railway gate crossings are critical points where railway tracks intersect with roadways, posing significant safety risks due to frequent interaction between trains, vehicles, and pedestrians. In many places, especially at unmanned or partially automated railway crossings, accidents still occur due to human negligence, delayed gate operation, or the absence of proper warning systems. Such incidents often lead to serious injuries and, in many cases, loss of life. These safety concerns highlight the need for a more reliable and responsive system to manage railway crossings.

To overcome these problems, automating railway gate control has become increasingly important. An automatic railway gate control system minimizes the need for manual operation and ensures that the gate opens and closes at the correct time based on the actual movement of trains. By continuously monitoring the presence of a train and controlling the gate accordingly, the system significantly improves safety and reduces the risk of accidents at railway crossings. This project involves the design and implementation of a simple and cost-effective automatic railway gate control system using embedded technology. IR sensors are used to detect the movement of the train, while an Arduino microcontroller processes the sensor signals and makes control decisions. A servo motor is employed to smoothly open and close the gate. By measuring the time taken for the train to pass between two sensors, the system is able to calculate the train's speed and intelligently manage the gate operation.

To ensure clear communication with road users, visual indicators such as LEDs and an LCD display are included to show real-time information about the gate status and train movement. The proposed system is especially suitable for unmanned or semi-automated railway crossings, where reliable and safe operation is critical. Overall, this project provides a practical and economical solution to a real-world transportation safety problem, helping to safeguard both road users and railway operations.

1.1 Motivation

While reading daily newspapers, it is often observed that many accidents occur at railway crossings.Railroad- related accidents are more dangerous than other transportation accidents in terms of severity and death rate etc.Therefore, greater efforts are required to improve railway safety. Trains carry enormous momentum compared to most other vehicles, and when accidents occur, they often result in serious damage to human life and property. Because of this, railway safety has become a major concern for rail operations worldwide. Ensuring safe train movement and preventing accidents are top priorities for railway authorities, who are constantly seeking better safety practices and advanced monitoring systems. One of the most serious safety challenges in railway systems is the development of cracks in railway tracks. If these cracks are not detected at an early stage, they can lead to severe accidents and derailments. Railway crossings also represent particularly dangerous zones, as they are unavoidable points where road and rail traffic meet. Even a small error or brief lapse in attention at these crossings can result in life-threatening accidents.A derailment occurs when a train or rail vehicle moves off its designated rails, although this does not always mean the train completely leaves the track. While some derailments may be minor, they still interrupt railway operations and pose safety risks. In certain emergency situations, controlled derailments using safety mechanisms such as catch points are intentionally used to prevent more serious accidents. These issues highlight the urgent need for reliable systems that can detect potential hazards early and help ensure safer and more efficient railway operations.

1.2 Objective of the project

- To design and develop an automatic railway level-crossing control system that functions reliably without the need for human intervention.
- To accurately detect the arrival and departure of trains using infrared (IR) sensors.

1.3 Methodology

The automatic railway crossing system works by continuously monitoring the railway track using sensors to detect the presence of a train. When a train approaches the cross-

ing, the sensor placed before the gate detects it and sends a signal to the microcontroller. The microcontroller processes this signal and automatically activates the motor driver to close the railway gate, while simultaneously turning on the buzzer and warning indicators to alert road users. After the train has completely passed the crossing, another sensor placed after the gate detects the train's departure and sends this information to the microcontroller. Based on this input, the controller deactivates the warning system and operates the motor in the reverse direction to open the gate. After completing its operation, the system switches back to standby mode, remaining alert to detect the next approaching train. This automated process reduces the need for human involvement and helps lower the risk of accidents, especially at unmanned railway crossings.

1.4 Organisation of the report

This project report is divided into five chapters. The first chapter provides an introduction to railway level crossings and explains the need for automation. It also discusses the problem being addressed, the objectives of the project, the motivation behind it, and the overall methodology followed. The second chapter presents a review of existing manual and automatic gate control systems, along with previous research related to accident prevention. The third chapter describes the complete system in detail, including the block diagram, working principle, hardware and software components, and their interconnections. Chapter four focuses on the results obtained from the system, evaluating its performance and reliability. Finally, the fifth chapter concludes the project and outlines possible future enhancements.

Chapter 2

Literature Survey

Unmanned Railway Level Crossings (UMLCs) in India pose a serious safety risk. These crossings allow road traffic and trains to intersect without protective gates, warning signals, or on-site personnel to manage traffic. Due to the lack of proper warning systems, poor visibility, and human errors by road users, many serious accidents occur at these crossings every year. Maintaining a safe distance and providing timely warnings to approaching vehicles are essential to prevent collisions with oncoming trains and to improve overall safety.

This project aims to develop an Information and Communication Technology (ICT)–based vehicular safe distance model specifically designed for Indian unmanned railway level crossings. The proposed approach makes use of ICT tools such as GPS, GIS mapping, traffic sensors, and data analytics to gather real-time and field-level information. Key factors such as vehicle speed, driver perception and reaction time, road geometry, and the physical characteristics of the crossing are analyzed to improve safety and provide timely warnings, ultimately helping to reduce accidents at these high-risk locations. . Using this data, a mathematical and statistical model is developed to estimate the minimum safe distance required for a vehicle to stop safely before reaching the railway track when a train is approaching.

The model primarily determines the optimal placement of traffic calming measures such as speed breakers and warning signs at appropriate distances from the crossing. Regression and sensitivity analysis techniques are used to identify the most influential parameters affecting safe distance, with special emphasis on driver reaction time and vehicle speed. The ICT framework enhances accuracy, repeatability, and adaptability compared to traditional empirical design methods.

The outcome of this project supports improved road safety planning at UMLCs by providing a scientifically validated approach for infrastructure design. It also lays the foundation for future intelligent transportation systems, such as real-time warning alerts and vehicle-to-infrastructure communication, contributing to accident reduction and safer rail-road

interaction in rural and semi-urban India. [1]

Railway level crossings are critical points where road and rail traffic intersect, and they are often prone to accidents due to human error, delayed gate operation, and the absence of monitoring at unmanned crossings.To improve safety and minimize accidents at railway crossings, this project proposes an automated railway crossing and obstacle detection system based on embedded systems and sensor technologies.The main goal of the system is to reduce human involvement while ensuring reliable and timely gate operation.

The system automatically manages the opening and closing of the railway gate by detecting the arrival and departure of a train using sensors placed on both sides of the level crossing. When a train approaches the crossing, the system activates warning signals and closes the gate automatically, ensuring that vehicles and pedestrians are prevented from entering the track area. After the train has safely passed the crossing, the gate opens automatically, allowing normal road traffic to continue without interruption.

In addition to automatic gate control, the project incorporates an obstacle detection feature to further enhance safety. Sensors such as ultrasonic or infrared sensors continuously monitor the crossing area to identify any vehicle, object, or person that may be trapped on the tracks during gate operation. If an obstacle is detected, the system immediately generates an alert to the train driver or the control station. This enables timely actions, such as slowing down or stopping the train, to avoid potential collisions. By combining automatic gate operation with obstacle detection, the system offers a more reliable and effective solution for improving safety at railway level crossings. [2]

Railway level crossings are major traffic bottlenecks and accident-prone zones, especially in urban and semi-urban areas where road traffic density is high. Conventional railway crossing systems rely heavily on manual control or fixed-time signaling, which often leads to traffic congestion, delays, and unsafe situations. To overcome these challenges, this project proposes a Smart Traffic Management System at Railway Crossing based on the Internet of Things (IoT).

The proposed system makes use of IoT-enabled sensors, microcontrollers, and wireless communication to continuously monitor both train movement and road traffic conditions in real time. Sensors installed along the railway track detect the arrival and departure of trains, while traffic sensors placed near the crossing measure vehicle density on the road. This real-time information is processed by an IoT controller, which intelligently controls

the opening and closing of the railway gate and manages traffic signals to improve overall traffic flow.

When a train is approaching the crossing, the system automatically turns on warning signals, closes the gate, and adjusts nearby traffic lights to prevent traffic congestion. After the train has safely passed, the gate opens automatically, and the traffic signals are regulated based on the current vehicle density, allowing vehicles to move smoothly and efficiently. In addition, all system data is sent to a cloud platform, enabling remote monitoring, data storage, and analysis through a web or mobile interface, making the system more efficient and easier to manage. [3]

Railway track monitoring is a critical aspect of ensuring safe and reliable train operations. Traditional railway track inspection methods depend heavily on manual checking, which is time-consuming and often affected by human error. Because of this, track defects may go unnoticed in their early stages, increasing the risk of accidents and service disruptions. To overcome these limitations, this project proposes a sustainable automated railway track detection system that uses smart sensing, automation, and energy-efficient technologies to improve reliability and safety.

The proposed system continuously monitors the condition of railway tracks using a combination of sensors such as ultrasonic, vibration, infrared, and camera-based modules. These sensors can be placed along the railway track or mounted on inspection vehicles to detect problems like cracks, track misalignment, breaks, or obstacles on the rails. An embedded controller collects and processes the sensor data, identifies any abnormal conditions, and generates real-time alerts so that necessary corrective actions can be taken promptly.

Sustainability is an important aspect of this project. The system operates using renewable energy sources such as solar power and employs low-power electronic components, which helps reduce energy consumption and maintenance requirements. Wireless communication through IoT modules enables track condition data to be transmitted to a central monitoring station or cloud platform for remote supervision and analysis. This allows for predictive maintenance, enhances railway safety, and supports reliable, energy-efficient, long-term operation. [4] The proposed system is designed to automatically control railway gates at unmThe proposed system is designed for unmanned railway level crossings to help prevent accidents caused by human error and negligence. It uses object-detection

sensors to identify the presence of trains and operates on solar power to ensure continuous and environmentally friendly functioning.

Sensors such as infrared or ultrasonic sensors are installed on both sides of the railway track at a fixed distance from the crossing. When a train approaches, the first sensor detects it and sends a signal to the microcontroller. Upon receiving this signal, the controller activates the motor mechanism to automatically close the railway gate. At the same time, warning indicators such as LEDs and an audible alarm are triggered to alert road users and prevent them from entering the crossing.

Once the train has completely passed the crossing, the second sensor detects its departure. The microcontroller then processes this information and commands the motor to reopen the gate, allowing normal road traffic to flow smoothly.

The entire system is powered by a solar panel connected to a rechargeable battery. The battery stores energy generated during the day, ensuring reliable operation even in remote or rural areas where grid power is unavailable. This makes the system energy-efficient, low in maintenance, and well suited for unmanned railway level crossings.

Overall, the project aims to enhance safety at railway crossings by automating gate operations, reducing dependence on manual control, minimizing accidents, and promoting the use of renewable energy sources. [5] This work focuses on customizing the Raspbian (Raspberry Pi OS) Linux kernel to support a dependable and real-time automatic railway level crossing system. The primary objective of this work is to improve overall system performance, hardware interaction, and response time so that safety-critical gate control operations can be carried out smoothly and reliably.

The development process begins with configuring the Raspbian kernel to support the required hardware drivers and kernel modules for sensors, motor drivers, GPIO pins, and communication interfaces. At the same time, unnecessary kernel components are removed to reduce system load, boot time, and power consumption. This optimization makes the platform more efficient and better suited for deployment at railway level crossings.

To achieve fast and accurate interaction with the hardware, custom kernel drivers are developed or modified for components such as IR or ultrasonic object-detection sensors, gate motor controllers, alarms, and visual indicators. Special attention is given to interrupt handling and task scheduling to ensure that approaching trains are detected promptly and that the gate responds immediately—an essential requirement for maintaining safety.

The customized kernel also incorporates real-time features, including priority-based task scheduling and faster GPIO access, to ensure predictable and consistent system behavior. In addition, power management settings are carefully optimized to support solar-powered operation, helping to extend battery life and maintain continuous operation in remote and unmanned locations.

Overall, this work results in a robust and optimized Raspbian kernel specifically designed for an automatic railway level crossing system. It enhances reliability, safety, and responsiveness, making the system well suited for long-term deployment in real-world railway environments. . [6]

Chapter 3

System Overview

The system is an automated railway level-crossing gate that uses multiple IR sensors and a servo-driven barrier to manage train movement and road traffic safely. When powered on, a microcontroller initializes the sensors, LCD display, and servo motor, keeping the gate open while continuously watching the track for an approaching train at the first sensor location. As the train passes successive sensors, the controller measures timing between detections to estimate speed and distance, then decides when to lower the gate so vehicles have cleared the crossing before the train arrives. Once a final sensor placed beyond the crossing confirms the train has fully passed, the controller raises the barrier back to its open position and resumes monitoring, providing an automatic, repeatable cycle with minimal human intervention and improved safety at the level crossing.

3.1 Flow Diagram of the model

The figure 3.1 illustrates how an automatic railway level-crossing gate operates using IR sensors and a servo motor to improve safety at the crossing. At the beginning, the controller powers up, initializes the sensors, LCD, and servo, and keeps the gate in the open position while continuously monitoring the first sensor near the track for an approaching train. When this sensor detects a train, the system notes the detection time and, together with the reading from a second sensor placed further along the track, estimates the train's speed and calculates how long to wait before closing the gate so that vehicles can safely clear the crossing. After the train reaches the second sensor, the controller concludes that the train is close enough to be a hazard, then issues a command to the servo motor to rotate and bring the barrier down, fully closing the gate before the train reaches the road. A third sensor beyond the crossing checks when the train has completely passed the danger zone; once this sensor is triggered, the controller again records the event, instructs the servo to rotate the gate back to the open position, and returns to its initial monitoring state, ready to manage the next train while keeping both rail and road users safe.

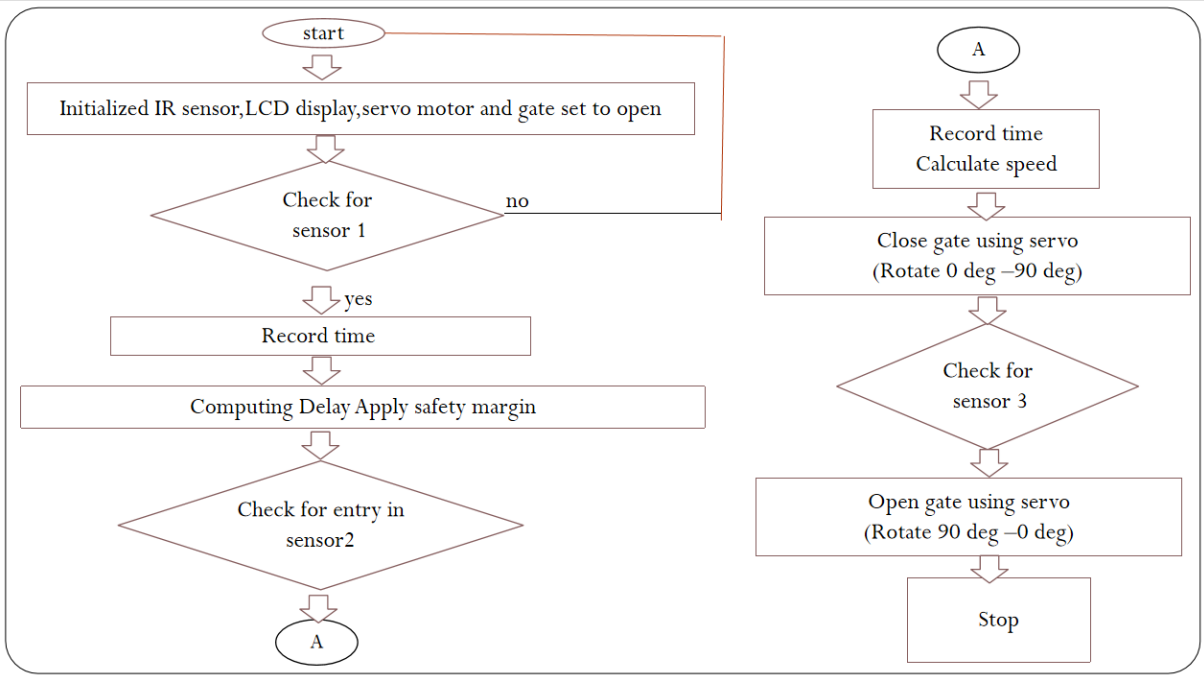

Figure 3.1: Flow graph of the system

3.2 System Hardware

The system hardware of the automatic railway crossing consists of various electronic and electromechanical components thaThe system works together to detect the train, control the gate, and warn road users.At the heart of the hardware setup is the Arduino Uno microcontroller, which serves as the main control unit of the system. It collects input signals from various sensors, processes the information, and generates appropriate control signals for the output devices. Infrared (IR) sensors are installed on both sides of the railway track to detect the arrival and departure of trains. These sensors continuously transmit and receive infrared signals, and when a train passes through the sensing area and interrupts the beam, the sensor output changes and sends a signal to the microcontroller. Proper placement and calibration of these sensors are essential for accurate train detection.

Since the microcontroller cannot supply sufficient power to drive the motor directly, a motor driver is used to amplify the control signals and provide the required current to the servo motor. The servo motor then smoothly opens and closes the railway gate based on the commands received from the microcontroller.

To enhance safety and provide clear alerts, LEDs are included in the hardware setup to indicate the gate status, such as when the gate is closing or remains closed. In addition, a

16×2 LCD display is used to show real-time messages like train arrival, gate closed, and gate open. These visual indicators help road users clearly understand the current status of the crossing.

Overall, the hardware design provides a reliable and efficient solution for automatic railway gate operation. It reduces dependence on manual control and significantly improves safety at unmanned railway crossings. The modular nature of the system also allows for easy future upgrades, such as integrating GSM modules, cameras, or IoT-based sensors.

3.2.1 Microcontroller

Microcontroller (Arduino Uno) The microcontroller acts as the brain of the automatic railway crossing system. It receives input signals from the sensors, processes them according to the programmed logic, and sends appropriate control commands to components such as the servo motor, LEDs, and the LCD display. The Arduino Uno is chosen for this project because it is easy to program, cost-effective, and provides sufficient input and output pins. These features make it well suited for handling the real-time control and automation required in railway crossing applications.

3.2.2 IR Sensors

IR Sensors The microcontroller acts as the central control unit of the automatic railway crossing system. It receives signals from the sensors, processes them according to the programmed logic, and then generates appropriate control signals to operate components such as the servo motor, LEDs, and the LCD display. The Arduino Uno is chosen for this application because it is easy to program, economical, and provides multiple input and output pins. These features make it well suited for real-time control and automation tasks required in railway crossing systems.

3.2.3 Servo Motor

Servo Motor The servo motor provides precise control over angular movement, which is essential for smooth and accurate gate operation. It is driven through a motor driver based on commands from the microcontroller, ensuring that the railway gate opens and closes safely and at the correct time.

3.2.4 LED

LED IndicatorsLED indicators provide clear visual signals showing the current status of the gate. They help improve safety by alerting pedestrians and drivers, ensuring they are aware when the gate is closing, closed, or opening.

3.2.5 LCD

LCD Display (16×2) The LCD display shows real-time messages like “Train Arriving,” “Gate Closed,” and “Gate Open,” keeping users informed about the current status of the crossing. This helps improve safety and awareness for both pedestrians and drivers.

3.2.6 Software Requirements

The software requirements define the tools and libraries needed to implement and execute the automatic railway gate control system. The system is developed using an embedded software environment that supports real-time control and hardware interfacing. Software Platform:

Arduino Integrated Development Environment (IDE)

Programming Language:

Embedded C / C++

Libraries Used:

Servo.h – for controlling the servo motor used as the railway gate

LiquidCrystal.h – for interfacing the 16×2 LCD display

Operating System:

Windows / Linux (any system supporting Arduino IDE)

Additional Tools:

Arduino Serial Monitor for debugging and monitoring system operation

The selected software tools provide reliable real-time performance, easy sensor interfacing, and low development complexity, making them suitable for safety-critical embedded applications.

3.2.7 Algorithm

The algorithm defines the logical sequence of operations carried out by the system software to ensure safe and efficient control of the railway gate.

Step 1: Initialize IR sensors, servo motor, LCD display, LEDs, and serial communication.

Step 2: Keep the gate in open position and display system status on LCD.

Step 3: When Sensor 1 detects the train, record the start time.

Step 4: When Sensor 2 detects the train, record the end time.

Step 5: Calculate the time difference and determine the train speed using the known distance between sensors.

Step 6: Compute the delay required for gate closure based on train speed and apply a safety margin.

Step 7: Close the gate using the servo motor and activate the red LED.

Step 8: Continuously monitor Sensor 3 to detect train exit.

Step 9: After the train passes Sensor 3, open the gate and activate the green LED.

Step 10: Display gate status on the LCD and return to monitoring mode.

3.2.8 Flowchart

The flowchart (refer figure 3.1) represents the operational sequence of the automatic railway gate control system. The process starts with system initialization, where all components—including the sensors, LCD display, servo motor, and LEDs—are set up and configured. Initially, the gate remains open to allow normal road traffic to pass freely.

The system then continuously monitors the entry sensors for any approaching train. When the train is detected by Sensor 1, the system records the start time. As the train passes Sensor 2, the end time is recorded, and the train's speed is calculated based on the time taken to travel between the two sensors and the known distance separating them. Using this speed, the system applies an appropriate delay before closing the gate, ensuring safe operation.

Once the delay is complete, the servo motor closes the gate, while the LEDs and LCD display provide visual and textual warnings to inform road users. The system continues to monitor the train's passage, detecting when it has fully cleared the crossing using Sensor

3. After the train passes, the gate automatically reopens, and the system returns to its initial state, ready to monitor for the next approaching train.

Chapter 4

Results

The automatic railway gate control system was successfully designed, implemented, and tested using Arduino, IR sensors, a servo motor, LCD display, and LED indicators. The system demonstrated reliable performance in detecting train movement, calculating train speed, and controlling the railway gate automatically without human intervention. The complete model is as shown in figure 4.1. During testing, the IR sensors accurately detected the presence of the train at different points along the track. Sensor 1 and Sensor 2 were used to measure the time taken by the train to travel a known distance, enabling precise calculation of train speed. The calculated speed was found to be consistent and suitable for determining the appropriate delay required for closing the gate. This adaptive timing ensured that the gate closed safely before the train reached the crossing, regardless of whether the train was moving at high or low speed.This explained in figure 4.2 The servo motor responded correctly to control signals from the microcontroller and performed smooth opening and closing of the gate. The gate closure was completed within the predefined time, ensuring safety for both road users and the train. After the train passed the crossing, Sensor 3 effectively detected the train's exit, and the gate was reopened automatically. This eliminated unnecessary delays for road traffic. The LCD display provided clear and real-time information regarding the gate status, such as "Gate is Open," "Gate is Closing," and "Gate is Closed."This shown in figure 4.4 The LED indicators further enhanced visibility, with the red LED indicating a closed gate and the green LED indicating an open gate..this shown in figure 4.3 These visual alerts improved user awareness and system transparency. The system operated continuously without failure during repeated trials, demonstrating good reliability and stability. The use of simple and low-cost components makes the system economically feasible for deployment at unmanned or semi-automated railway crossings. Overall, the results confirm that the proposed automatic railway gate control system is effective in improving safety, reducing human involvement, and ensuring efficient traffic management at railway level crossings.

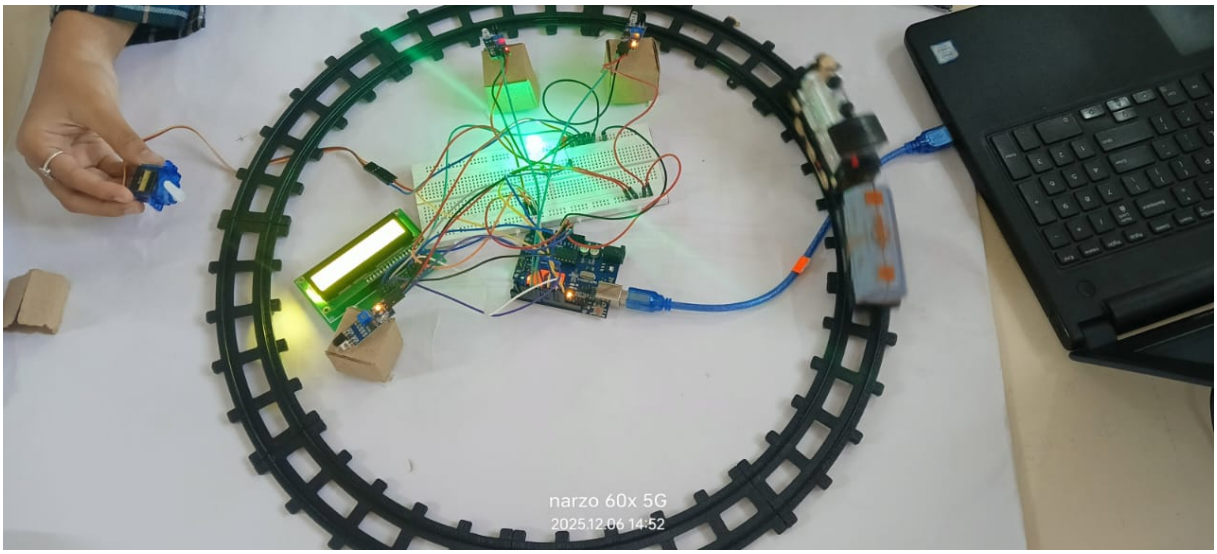

Figure 4.1: Complete model of railway crossing system

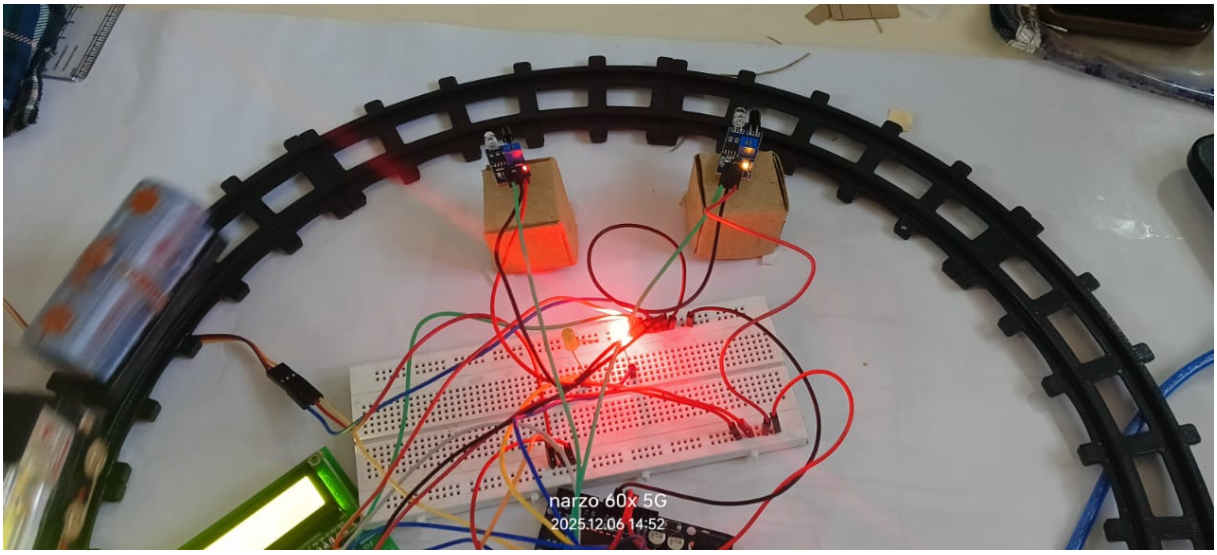

Figure 4.2: Representing train is arriving towards the gate and gate is closed

Indicates the train is passed

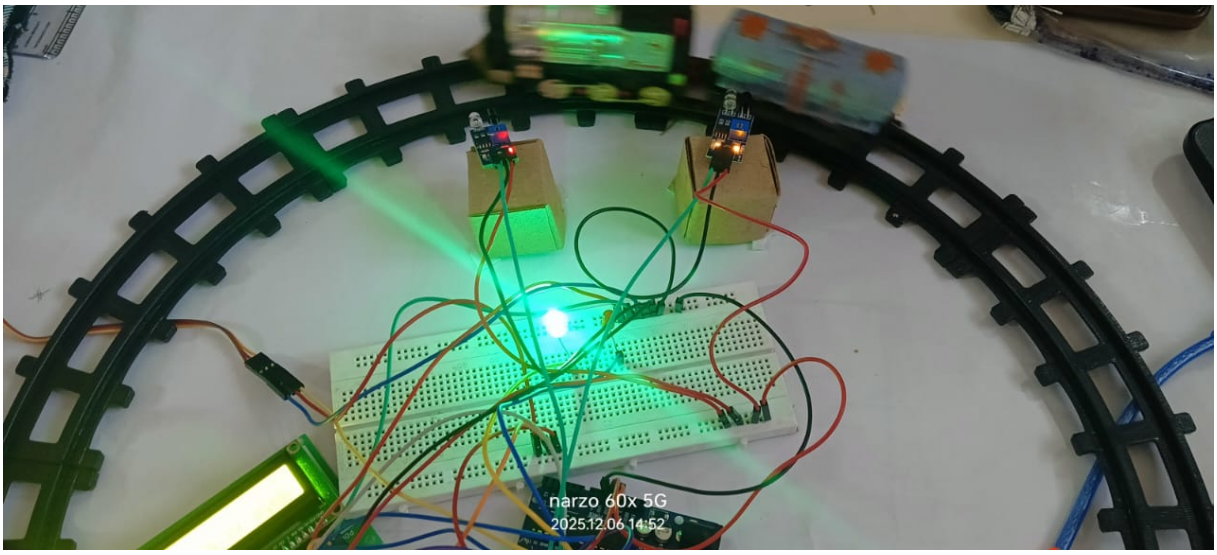

Figure 4.3: Indicates the train is passed

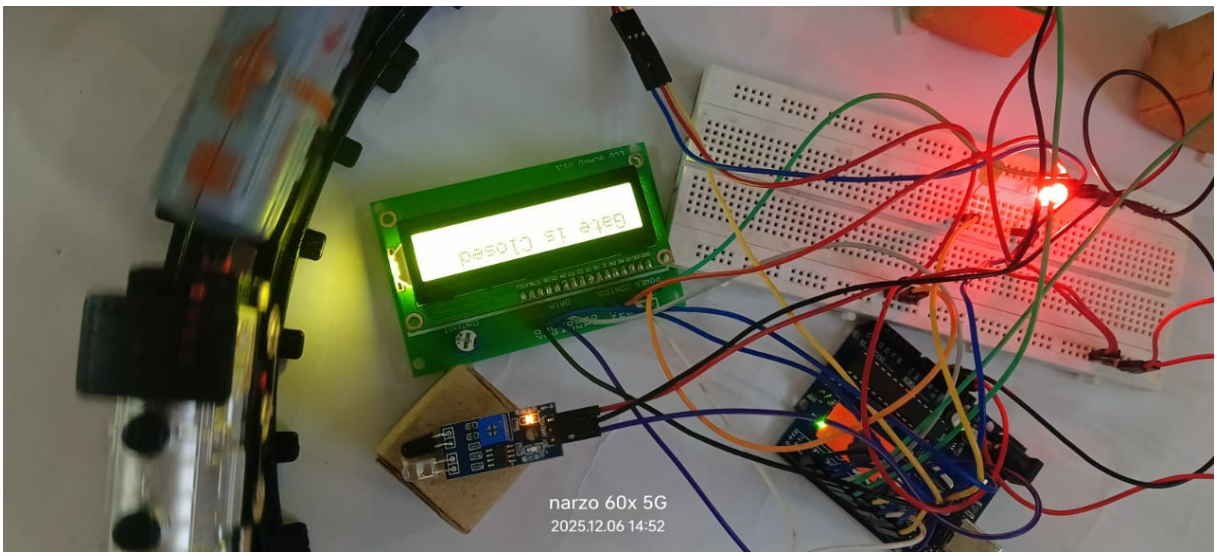

Figure 4.4: Display indication about gate closed

Chapter 5

Conclusion

The automatic railway crossing system developed in this project demonstrates a reliable and safe way to control railway gates without the need for human intervention. By automatically detecting approaching trains and adjusting the gate operation based on their speed, the system ensures that the gate closes at the correct time for both slow-moving and fast-moving trains. This timely operation greatly reduces the risk of accidents caused by delayed or incorrect gate control.

The system also provides clear visual indicators and real-time status updates, keeping road users informed and improving overall safety at the crossing. Once the train has safely passed, the gate reopens automatically, minimizing unnecessary waiting time for vehicles. The observed results show that the gate operates smoothly and accurately, confirming the effectiveness of the implemented control logic.

Overall, the system offers a cost-effective and efficient solution for improving safety at railway crossings, particularly in unmanned or remote locations. With further enhancements, it can be integrated into larger real-time railway safety and traffic management systems, contributing to improved public safety and more efficient railway operations.

5.1 Scope for future work

The automatic railway crossing system has great potential for further improvement and wider use.Several enhancements can be made to further improve the effectiveness of the proposed system. One possible improvement is connecting the system to the internet and cloud platforms, which would enable real-time monitoring, remote control, and continuous status updates to a central control center.

Accuracy can be further improved by using advanced sensors such as LIDAR or ultrasonic sensors to measure train speed and distance more precisely. This would help ensure that the gates open and close at the most appropriate times. Additional emergency management features can also be incorporated to detect situations such as stalled vehicles

or obstacles on the tracks, automatically triggering alarms and instantly alerting the relevant authorities.

For unmanned or remote railway crossings, powering the system using solar energy would make it more sustainable and reliable, especially in areas with limited or no access to conventional power supply. Integration with nearby traffic signals could also help manage road traffic more efficiently while maintaining railway safety.

In the future, artificial intelligence techniques could be used to predict train arrivals and dynamically adjust gate operations in real time. This would help reduce unnecessary waiting time for vehicles and improve overall system efficiency. Finally, the system can be extended to support multiple tracks or more complex crossing layouts, making it adaptable to a wider range of railway environments.

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Appendices

Appendix A

Sustainable Development Goals addressed

Table 1.1: Sustainable Development Goals of the Project

#	**SDG**	**Level**
1	No Poverty	-
2	Zero Hunger	-
3	Good Health and Well-being	2
4	Quality education	-
5	Gender Quality	-
6	Clean water and Sanitation	-
7	Affordable and Clean Energy	1
8	Decent work and Economic Growth	-
9	Industry, Innovation and Infrastructure	1
10	Reduced Inequalities	-
11	Sustainable cities and Communities	-
12	Responsible Consumption and production	-
13	Climate action	-
14	Life below water	-
15	Life on Land	-
16	Peace, Justice and Strong Institutions	-
17	Partnership's for the Goals	-

Levels: Poor:1, Good :2, Excellent:3

Appendix B

Self-Assessment of the Project

Sl. No.	PO PSO	Contribution from the Project	Level
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.	MATLAB script for analyzing hybrid beamforming, including SNR-based capacity, frequency response, interference modeling, directivity patterns, and OFDM constellation diagrams.	2
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)	Identified challenges in analog and digital beamforming so developed a practical analysis of hybrid beamforming	2

PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5) .	Designed a hybrid beamforming techniques for same frequency and different distances and diffrent frequencies for same distance with the combining features of analog and digital beamforming	2
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions. (WK8).	Implementing hybrid beamforming simulations and investigating its performance under various conditions.	2
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).	A comprehensive MATLAB script related to hybrid beamforming.	2
PO6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).	The script is well-structured and addresses different aspects of beamforming systems, including interference, directivity, and frequency response	2

PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws. (WK9)	Followed ethical considerations in technology use.	3
PO8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multidisciplinary teams.	Collaborated with peers, integrating MATLAB software efficiently.	3
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences	Documented processes and communicated technical concepts clearly during implementation and presentation.	2
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.	Managed resources effectively, optimizing interferences and beamforming costs.	2
PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)	Demonstrated adaptability by integrating new tools and handling emerging challenges.	2

PSO1	Apply and analyze the concepts of circuits and systems for real time challenges in the areas of Electronic Circuits, Signal Processing and VLSI/Embedded Systems.	Applied concepts of signal processing that has several fundamental concepts to control the direction and shape of the radiated signal.	2
PSO2	Identify, design and develop solutions for complex engineering problems related to, Communication Systems using analytical techniques, state of the art simulation tools and hardware.	Designed a solution for analog and digital beamforming challenges.	3

Levels: Poor:1, Good :2, Excellent:3

Appendix C

Data Sheet of IR sensor

- High-sensitivity infrared presence and motion detection sensor
- Reach up to 4 meters without lens for objects measuring 70 x 25 cm^2
- Integrated silicon IR filter
- SMD friendly
- Capable of detecting stationary objects
- Capable of distinguishing between stationary and moving objects
- 80° field of view
- Factory calibrated
- Low power
- Embedded smart algorithm for presence / motion detection

Electrical specifications

- Supply voltage: 1.7 V to 3.6 V
- Supply current: 10 µA
- 2-wire I^2C / 3-wire SPI serial interface
- Programmable ODRs from 0.25 Hz to 30 Hz
- One-shot mode Sensing specifications
- IR sensitivity: 2000 LSB/°C
- RMS noise: 25 LSBrms
- Operating wavelength: 5 µm to 20 µm
- Local temperature sensor accuracy: ±0.3

Appendix D

Data Sheet of LCD

- Type: Character
- Display format: 16 x 2 characters
- Built-in controller: ST 7066 (or equivalent)
- Duty cycle: 1/16
- 5 x 8 dots includes cursor
- + 5 V power supply
- LED can be driven by0 pin 1, pin 2, or A and K
- N.V. optional for + 3 V power supply
- Optional: Smaller character size (2.95 mm x 4.35 mm)